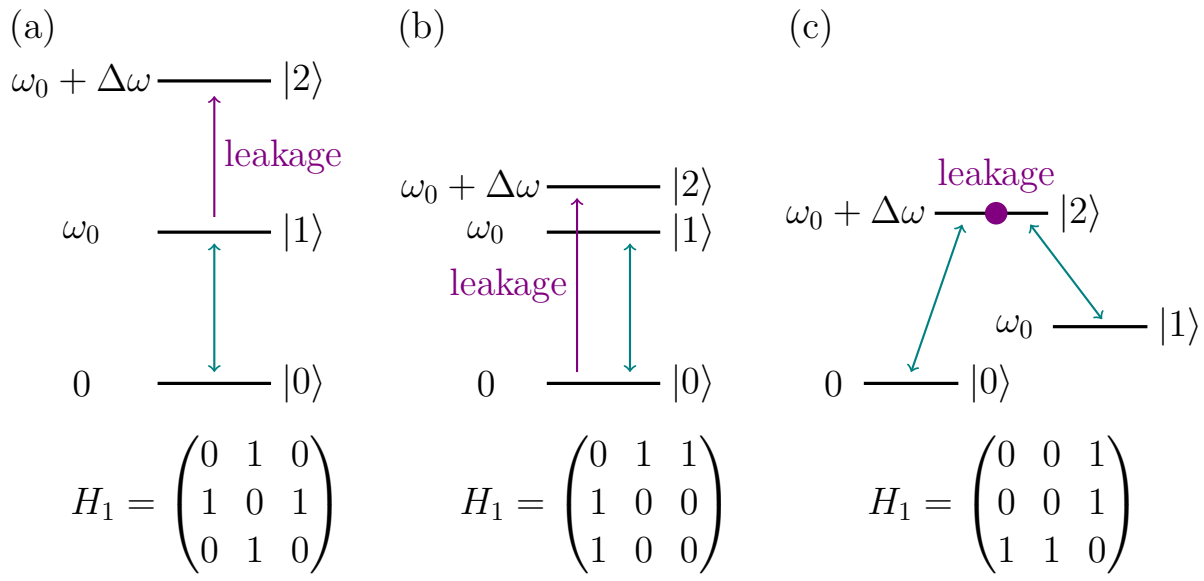
(a)
$\omega_0 + \Delta\omega$
$|2\rangle$
leakage
$\omega_0$
$|1\rangle$
0
$|0\rangle$
$H_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$
(b)
$\omega_0 + \Delta\omega$
$|2\rangle$
$\omega_0$
$|1\rangle$
leakage
0
$|0\rangle$
$H_1 = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$
(c)
leakage
$\omega_0 + \Delta\omega$
$|2\rangle$
$\omega_0$
$|1\rangle$
0
$|0\rangle$
$H_1 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$